

ELATMA, Russian Federation

WMO: 276480

Lat: 54.9722N

Lon: 41.7460E

Elev: 136

StdP: 99.71

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.6	-22.7	-28.2	0.3	-25.2	-25.3	0.4	-22.4	7.5	-5.0	6.8	-4.5	1.3	260	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	30.6	20.4	28.6	19.6	26.8	19.0	21.6	28.4	20.7	26.9	19.7	25.1	2.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	14.4	24.6	18.5	13.6	23.5	17.7	12.9	22.5	63.5	28.2	60.4	27.2	57.0	25.1	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	5.9	5.2	DB	-29.6	32.9	3.7	2.0	-32.3	34.4	-34.4	35.6	-36.4	36.7	-39.1	38.2	(4)
(5)			WB	-29.6	23.3	3.5	1.2	-32.2	24.1	-34.2	24.8	-36.2	25.5	-38.7	26.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-8.9	-8.0	-2.6	6.4	13.6	17.1	19.8	17.7	12.0	5.3	-1.2	-6.3	(6)	
	DBStd	11.39	7.03	6.92	4.75	5.15	4.94	4.06	3.59	3.98	4.04	4.37	5.24	6.60	(7)	
	HDD10.0	2659	586	505	391	133	21	1	0	0	25	156	336	505	(8)	
	HDD18.3	4856	844	738	649	359	161	71	26	60	195	404	586	763	(9)	
	CDD10.0	1008	0	0	0	24	133	215	304	239	83	9	0	0	(10)	
	CDD18.3	163	0	0	0	0	14	34	72	41	3	0	0	0	(11)	
	CDH23.3	1477	0	0	0	2	142	303	623	378	28	0	0	0	(12)	
	CDH26.7	441	0	0	0	0	31	81	198	127	3	0	0	0	(13)	
(14) Wind	WSAvg	2.4	2.6	2.6	2.6	2.5	2.3	2.2	1.8	1.9	2.0	2.5	2.7	2.7	(14)	
(15) Precipitation	PrecAvg	605	43	37	29	35	41	57	88	64	53	56	55	49	(15)	
	PrecMax	880	84	80	62	82	178	161	218	192	133	160	111	105	(16)	
	PrecMin	378	1	0	9	6	1	5	10	0	5	0	10	3	(17)	
	PrecStd	120	21	20	14	17	36	33	50	39	32	32	23	24	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	3.0	11.7	22.9	29.5	31.7	33.9	33.9	26.4	18.5	11.3	6.1	(19)	
		MCWB	2.0	1.6	7.1	13.2	18.8	19.9	22.0	21.9	17.4	13.1	9.7	5.5	(20)	
	2%	DB	1.6	2.0	7.7	20.1	27.0	28.7	30.7	30.1	23.2	15.3	8.5	3.9	(21)	
		MCWB	0.8	0.9	3.9	12.1	17.3	19.2	20.5	20.9	16.4	12.1	7.4	3.1	(22)	
	5%	DB	0.8	1.2	5.2	17.6	24.4	26.9	28.8	27.2	20.8	13.2	6.8	2.3	(23)	
		MCWB	0.3	0.5	2.5	11.0	16.2	19.0	20.2	19.2	15.2	10.6	5.7	1.6	(24)	
	10%	DB	0.1	0.6	3.4	15.2	22.1	24.7	26.9	24.9	18.5	11.5	5.0	1.1	(25)	
		MCWB	-0.4	-0.1	1.4	9.5	14.8	18.0	19.7	18.1	14.1	9.5	4.3	0.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.4	2.2	7.5	14.2	19.8	21.7	23.4	22.6	18.7	14.2	9.6	5.6	(27)	
		MCDB	2.7	2.5	11.4	20.8	27.6	28.2	30.7	31.5	24.2	16.8	11.0	6.0	(28)	
	2%	WB	1.1	1.2	4.3	12.9	18.1	20.7	22.0	21.2	17.3	12.5	7.7	3.3	(29)	
		MCDB	1.4	1.6	6.9	18.8	25.1	26.9	28.6	28.8	21.9	14.8	8.4	3.8	(30)	
	5%	WB	0.4	0.6	2.7	11.5	17.0	19.6	21.1	20.0	15.9	11.0	5.9	1.7	(31)	
		MCDB	0.8	1.1	4.6	16.7	23.4	25.2	27.1	25.8	19.5	12.9	6.7	2.1	(32)	
	10%	WB	-0.4	0.0	1.7	10.0	15.8	18.6	20.2	18.8	14.6	9.7	4.3	0.6	(33)	
		MCDB	0.1	0.7	3.1	14.5	21.4	23.5	25.4	23.6	18.0	11.2	4.9	1.0	(34)	
(35) Mean Daily Temperature Range	MDBR		5.7	6.7	7.6	9.5	10.7	9.9	10.3	10.2	8.9	6.1	4.4	4.8	(35)	
	5% DB	MCDBR	3.9	4.7	9.1	12.8	13.3	12.9	13.2	13.9	12.4	8.7	6.0	3.7	(36)	
		MCWBR	3.9	4.2	6.0	6.6	6.1	5.5	5.4	5.7	6.5	5.8	5.0	3.6	(37)	
	5% WB	MCDBR	3.9	4.4	7.7	11.7	12.3	11.4	11.3	12.2	10.8	7.8	5.4	3.7	(38)	
		MCWBR	3.9	4.1	5.4	6.5	6.2	5.4	5.3	5.6	6.3	5.8	4.9	3.7	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.289	0.328	0.355	0.416	0.388	0.382	0.412	0.431	0.378	0.358	0.322	0.291	(40)	
	taud		2.095	2.040	2.051	2.120	2.329	2.370	2.301	2.231	2.360	2.326	2.221	2.096	(41)	
	Ebn at Noon		612	716	796	801	855	864	826	780	779	695	576	513	(42)	
	Edh at Noon		66	100	125	136	117	113	120	121	93	77	59	53	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.68	1.59	3.15	4.09	5.52	5.60	5.63	4.56	2.87	1.36	0.55	0.37	(44)	
	RadStd		0.14	0.24	0.35	0.35	0.44	0.57	0.53	0.47	0.42	0.20	0.12	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+0.74	+0.55	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		+0.67	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page